POWER MODULE

ARK 12S PAB Power Module

5.2 V / 6 A regulated supply and digital power monitor for the Pixhawk Autopilot Bus

The USA-built ARK 12S PAB Power Module powers a Pixhawk Autopilot Bus carrier and monitors its battery. A TI INA238 measures a 100 A continuous battery path through a 0.1 mΩ shunt, and a synchronous buck supplies 5.2 V at 6 A for the avionics.

Its 6-pin Molex CLIK-Mate output matches the ARK PAB Carrier power pinout.

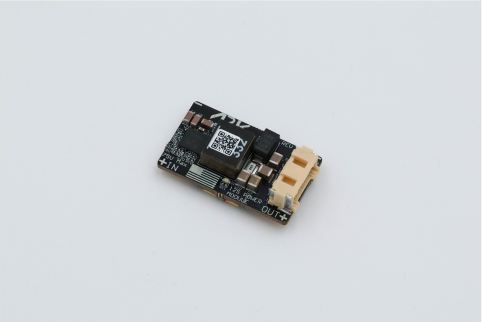
KEY FEATURES

- 100 A continuous battery current path (parallel for higher)
- TI LM70880 synchronous buck — 5.2 V / 6 A regulated output
- TI INA238 digital power monitor with 0.1 mΩ shunt (I²C)
- 10–75 V input range (12S capable)
- 6-pin Molex CLIK-Mate output matching PAB pinout
- Solder-pad battery input and output
- Includes 6-pin Molex cable, 12S capacitor and TVS diode

SPECIFICATIONS

ELECTRICAL	
Input voltage	10–75 V
Battery current	100 A continuous (20 °C ambient)
Regulated output	5.2 V @ 6 A continuous
Power monitor	INA238, 0.1 mΩ shunt, I²C address 0x45
Operating temperature	−40 °C to +85 °C
Storage temperature	−40 °C to +85 °C

MECHANICAL	
Dimensions	37 × 22 × 13 mm
Weight	11.5 g



AT A GLANCE	
Input	10–75 V
Battery current	100 A cont.
Output	5.2 V / 6 A
Monitor	INA238 0.1 mΩ
Connector	6-pin Molex
Dimensions	37 × 22 × 13 mm
Weight	11.5 g
Hardware rev	Rev 3.1

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-12s-pab-power-module/

DOCS docs.arkelectron.com/products/power/ark-12s-pab-power-module

CAD github.com/ARK-Electronics/ark-hardware

OUTPUT CONNECTOR PINOUT

PIN	SIGNAL	VOLTAGE	NOTES
1	VBRICK	5.2 V	Regulated output, 3 A per pin
2	VBRICK	5.2 V	Regulated output, 3 A per pin
3	SCL	3.3 V	I²C clock to INA238
4	SDA	3.3 V	I²C data to INA238
5	GND	GND	
6	GND	GND	

Connector: Molex 5024940670 6-position CLIK-Mate. Matches ARK PAB Carrier power pinout.

NOTES

- The power monitor measures only current flowing from the battery input pads to the battery output pads, so wire every battery load to the output pads.
- Active cooling is recommended when operating at high battery current or high 5 V regulator load.
- For continuous battery current above 100 A, operate multiple modules in parallel.
- When using 12S batteries, solder the included capacitor and TVS diode to the battery-output pads (observe polarity).

ARK-DS-ARK-12S-PAB-Power-Module | Version 1.0, 2026-10-29 | Compiled from the ARK 12S PAB Power Module Production schematic Rev 3.1 (24 Apr 2026), the Production bill of materials, the production STEP model, and ARK Electronics published documentation. Temperature ratings per component data sheets. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.